

Coding & Solution

Date	19 July 2026
Team ID	XXXXX
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/ishnu56/Voice-Based-Concept-Understanding-Analyser.git
Programming Language(s)	Python (Backend & ML), HTML5, CSS3, JavaScript (Frontend)
Framework(s) Used	Flask (Web Framework), OpenAI Whisper (Speech-to-Text), Sentence-BERT (Semantic Similarity), ReportLab (PDF Generation), SQLAlchemy (Database ORM)
Key Features Implemented	1. Speech-to-Text conversion using OpenAI Whisper. 2. Semantic similarity analysis using Sentence-BERT. 3. Multi-criteria scoring (grammar, pronunciation, fluency, comprehension). 4. Detailed performance reports and downloadable certificates. 5. User authentication and session management.
Pending / Incomplete Features	None. All core functional requirements, including speech processing, evaluation, report generation, and certificate creation have been successfully implemented and tested.
Setup / Run Instructions	1. Install dependencies: <code>pip install -r requirements.txt</code> 2. Set environment variables (e.g., <code>FLASK_APP=app.py</code>, <code>DATABASE_URL</code>). 3. Initialize database (if applicable). 4. Run application: <code>python app.py</code> 5. Access via browser at <code>http://127.0.0.1:5000</code>

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The provided solution effectively integrates speech recognition, semantic understanding, and multi-criteria evaluation to deliver accurate and insightful analysis. Flask is used for lightweight and efficient routing, while modular design and reusable components ensure scalability. The codebase follows PEP-8 standards and MVC architecture, ensuring readability, maintainability, and ease of future enhancements.